

Chapter 40

Text Based Editing

Recent advances in Machine Learning have made it possible to organize, edit, and refine clips by analyzing the spoken dialog, then converting that speech into text.

This chapter covers the usage of DaVinci Resolve's Text Based Editing toolset.

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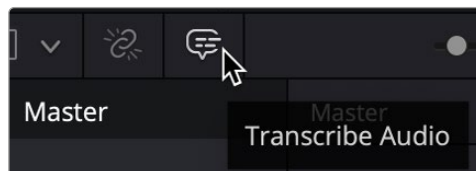
Audio Transcription (Studio Version Only)

The most critical metadata about any clip is knowing what people have said in it. A full transcription of a shot is useful in narrative film letting you find specific clips based on the dialog in the script, but a transcript is especially important in unscripted documentary and news production, both to understand what pieces of the story you actually captured and for a variety of organizational, creative and legal requirements.

Until recently, transcribing audio was a labor-intensive process requiring a human being to listen to a clip in real time and then type out what was being said in a log sheet. With recent advances in the DaVinci Resolve Neural Engine, your computer can now perform the tedious work of transcribing each clip for you automatically, and most importantly, accurately. In addition, having text transcripts attached to the clips in your project gives you powerful new text-based editing tools to select, search and insert clips into your timeline.

Transcribe Audio

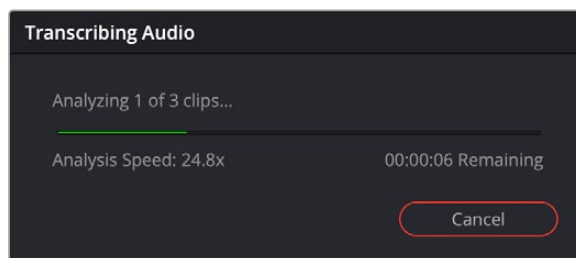
The Transcribe Audio feature is accessed via the Media Pool in the Media, Cut, Edit, and Fairlight pages and is a fully automated process.



The Transcribe Audio icon in the Media and Edit pages

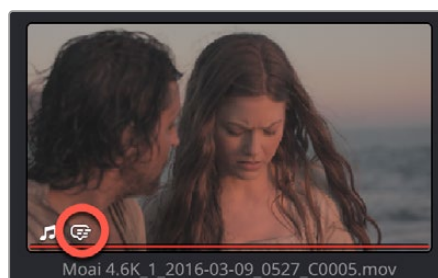
To automatically transcribe the audio of a clip:

- 1 Select a clip or clips with audio to transcribe in the Media Pool.
- 2 Right-click on any of the selected clips and select AI Tools > Audio Transcription > Transcribe from the contextual menu. Or click on the Transcribe Audio icon in the Media Pool toolbar.



The Transcribing Audio window

At this point the Transcribing Audio window will appear, showing you how many clips are left to analyze, the speed at which the analysis is taking place vs. real time, and approximately how much longer the analysis of the selected clips will take. There is a button to cancel this operation at any time.



The speech balloon icon showing that a clip has been transcribed

Any clip that has a transcription attached to it will have a small speech balloon icon appear in the lower left of its thumbnail in the Media Pool.

To remove a transcription from a clip:

- 1 Select a clip you want to delete the transcription from in the Media Pool.
- 2 Right-click on the selected clips and select AI Tools > Audio Transcription > Clear Transcription from the contextual menu.

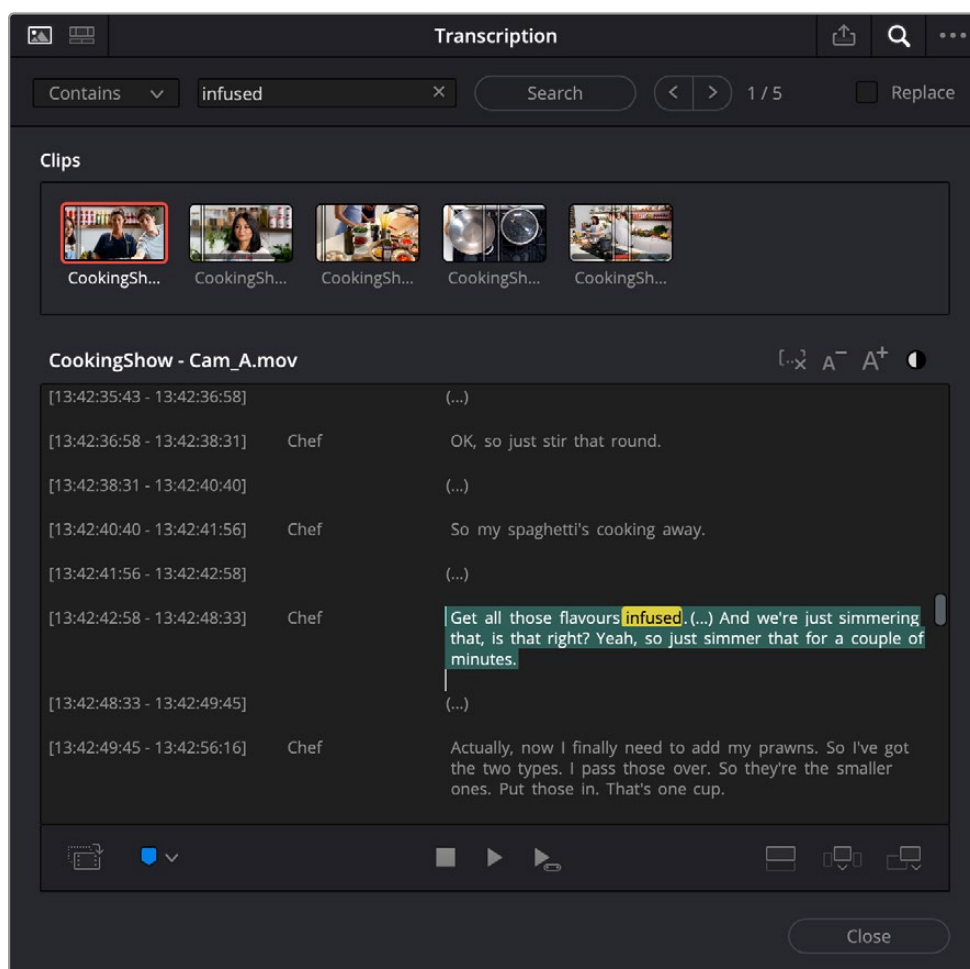
Audio Transcription of Timelines

Audio Transcription works not just with clips but for timelines as well and in the same way as described above. There are several practical uses for a text transcription of a timeline for legal and organizational requirements. However, if you're intending to make subtitles or captions for the Timeline, there is a dedicated tool specifically for that purpose.

For more information on the Create Subtitles from Audio feature, see *Chapter 52, "Subtitles and Closed Captioning."*

The Transcription Window

The Transcription window is where the text is displayed, along with its timecode, options, and editing tools. For ease of use the Transcription window is freely movable around the GUI, even to another monitor.



The Transcription window showing multiple clips with the Search pane open. The searched for text is highlighted in yellow. The In and Out range of the clip is highlighted in teal, and the current playhead position in the Viewer is indicated by a red triangle.

To access the Transcription window:

- Right-click on a clip or clips that have been transcribed, and select Transcribe Audio from the contextual menu. Or click on the Transcribe Audio icon in the Media Pool toolbar.

The Transcription window is comprised of the following sections and tools:

	Export: Clicking this icon will launch a file browser, where you can save a .txt file of the transcription results for use in other programs.
	Search: Clicking this icon will expose a Search and Replace dialog, letting you find specific words or phrases across all the selected clips in the Clips panel. Found words will be highlighted in yellow. You can also search and replace using partial words in the Transcription window. To change the parameters of the search, open the search bar and select either Contains or Full Word from the drop-down menu to the left.  You can also search for partial words in the Transcription window by using the Contains option.
	Options Menu: Reveals the Transcription window options. ▪ Speaker Detection: Enabling this option will analyze and assign a speaker number to a specific voice. This speaker is subsequently remembered across all clips in a project. ▪ Display Speakers: Turns the Speaker column on or off in the Transcription window. ▪ Transcribe Missing Clips in Timeline: If you've added any clips into the timeline that haven't yet been analyzed for speech to text (transcription pending), selecting this option will transcribe them and place them appropriately in the Transcription window. ▪ Remove Silent Portions: Removes all silences (...) detected from the Transcription window. ▪ Create Subtitles from Audio (Timeline only): Creates a new Subtitle track and places subtitles automatically. For more information on the Create Subtitles from Audio feature, see <i>Chapter 52, "Subtitles and Closed Captioning."</i> ▪ Import Subtitles (Clip only): Allows you to import a subtitle in .srt format (described later in this chapter). ▪ Export Subtitles: Allows you to export your subtitles in .srt format (described later in this chapter). ▪ Reset Speaker Data: Removes all saved speaker detection data from the project.
	Clips: This panel shows all the selected clips that you've chosen to see the transcription for. Clicking on a thumbnail loads that clip's transcription into the Transcript window. The selected clip is outlined in orange. If you only selected one clip, this panel does not appear. Transcript: Displays the transcription text. You can also use this box to select the playhead position, and set in and out points of a clip as described in the Text Based Editing section below.

	Remove Silent Portions: Removes all detected silences (...) from the transcript. Text View Controls: Allows you to control the font size and the high contrast settings of the text in the Transcript window. 
	Create Subclip: Creates a new subclip of the shot based on the In and Out points set by the highlight in the Transcript window.
	Marker: Places a marker at the word location that is highlighted in red, or if an In/Out range is highlighted, it will place a duration marker for the range instead.
	Stop: Stops playback of the clip in the Viewer.
	Play: Plays the clip forward realtime in the Viewer.
	Play in to Out: Plays forward only between the set In and Out range.
	Place on Top: Performs a Place on Top edit of the selected range into the timeline.
	Insert: Performs an Insert edit of the selected range into the timeline.
	Append: Performs an Append edit of the selected range to the end of the timeline.

Editing Text in the Transcription Window

Not everything transcribed in the audio of a clip is useful or wanted in the final transcription. Crew chatter, flubbed lines, and long silences can all be edited out, as well as mistakes fixed in the Audio Transcription process itself.

Selecting Text

In order to edit text, you first need to select it in the Transcription window. There are a few ways of doing so, depending on how much text you wish to select.

To select a specific word in the transcription:

Double-click on the word to select it and highlight it in teal.

To select a range of text in the transcription:

Click and drag over the text you want to edit to select it and highlight it in teal.

To select a sentence in the transcription:

Click on any word in the sentence and press the X key to select the whole sentence and highlight it in teal.

To select an entire paragraph in the transcription:

Triple-click on any word in the paragraph to select the whole paragraph and highlight it in teal.

To select all the text in the transcription:

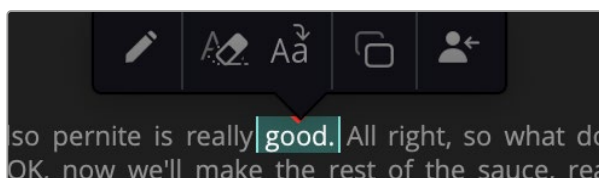
Press Command-A to select all of the text and highlight it in teal.

To deselect select all the text in the transcription:

Press Option-X to deselect the all of the text and highlight it in teal.

Editing Text

Once any text is selected (highlighted in teal), you can right-click on it to bring up the editing options. Clicking on any of the editing tools, executes the action.



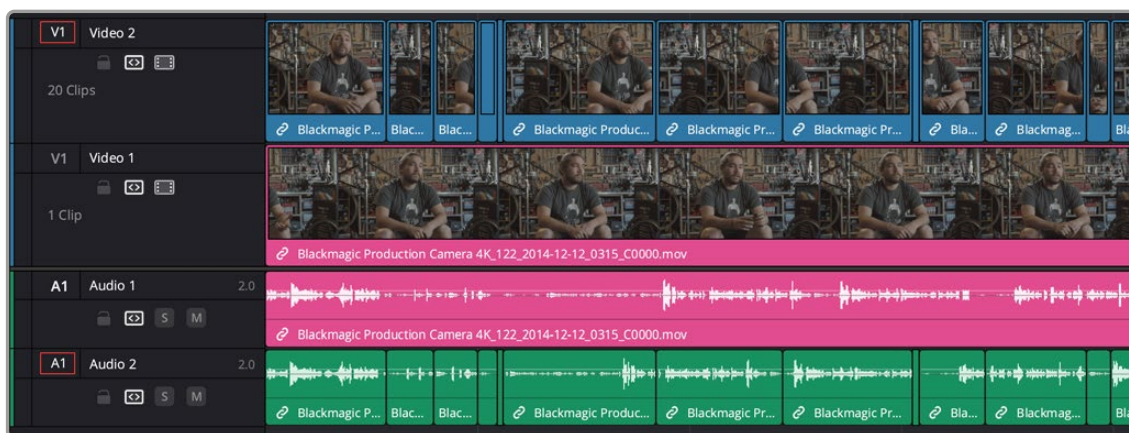
The Transcription window editing tools (l-r): Edit, Delete, Undelete, Copy, and Assign To, are accessed by right-clicking on any selected text.

	Edit: Clicking on this tool brings up a text field, where you can perform simple edits on the selected text. You can use this to correct the transcription in terms of wrong words, spelling mistakes and adjusting proper names. Radically altering the text, such as replacing an entire paragraph with a two or three word summary will have unpredictable effects if used for editing. This action is undoable.
	Delete: Clicking this tool does not delete the selected text from the Transcription window as in a word processor, instead it will mark the text in orange with a strike-through. Any text thus marked will not appear in the exported transcript, nor will its range be inserted into the Timeline with any edit operation. This means a cut will appear in the Timeline where the deleted text range is in the transcript. Since the words are all still there in the audio clip as they were recorded, it may be easier to think of this as an “exclude” operation rather than a deletion.
	Undelete: Clicking on this tool reverses any previous deletion. It changes the orange strike-through text back into normal text, allows it back into the exported transcription, and re-includes its range for editing purposes.
	Copy: Copies the selected range for use in other applications. There is no Paste command for the Transcription window.
	Assign To: If the automatic speaker detection has made an error in who is saying this line, you can manually assign it to a different speaker by selecting one from the drop-down menu.

Understanding Silence (...) in the Transcription Window

In most shots, a significant portion of the audio clip will have no dialog at all. These silences are marked in the Transcription window as ellipses (...). As important as knowing where the dialog is, it's also useful to know where it isn't. These silences (...) can be treated just like words in the Transcription window, meaning they can be deleted, and thus excluded from any editing operation.

For example, say you have a long interview with frequent pauses between questions. You could cut this long clip up manually by setting In and Out points and picking out the individual questions and answers. But, by using Audio Transcription and deleting all the silences (...) in the Transcription window and then editing that range into the Timeline, you would instantly have a series of individual clips laid down comprised of only the questions and the answers. This lets you automatically insert all of the relevant clips at once, without all the tedious scrubbing and setting of edit points for every single section of the clip in the Viewer.



The original Interview clip (Tracks V1, A1 violet) and the same media with all silences removed using the Transcription window (Tracks V2, A2). Note how it is automatically broken into individual question and answer clips.

Removing silences in the Transcript window has many useful applications for both exporting a transcript and in text based editing, and so has a dedicated tool for deleting them all.

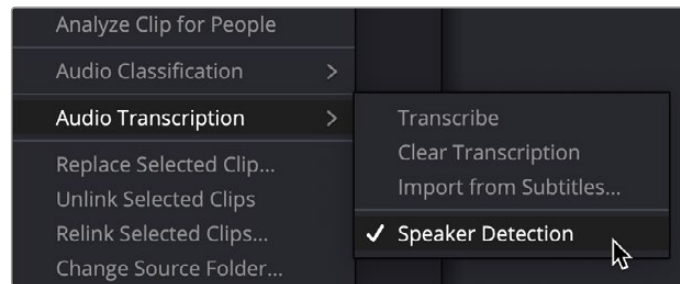
To delete all the silences (...) in the Transcription window:

- Click on the Remove Silent Portions icon just to the left of the Font Controls, in the Transcription window.
- Click on the Option menu (three dots) in the upper right of the Transcription window, and select Remove Silent Portions from the contextual menu.

Detect Speakers During Transcription (Studio Version Only)

DaVinci Resolve can now detect individual speakers when transcribing a clip, showing their specific timecode range as well as the text for each speaker. Speaker detection can not only create more useful text transcription exports but can be used for text based editing references.

Speaker Detection is project specific, so once you turn it on, it will remain on for all further transcription operations, either for clips or timelines.



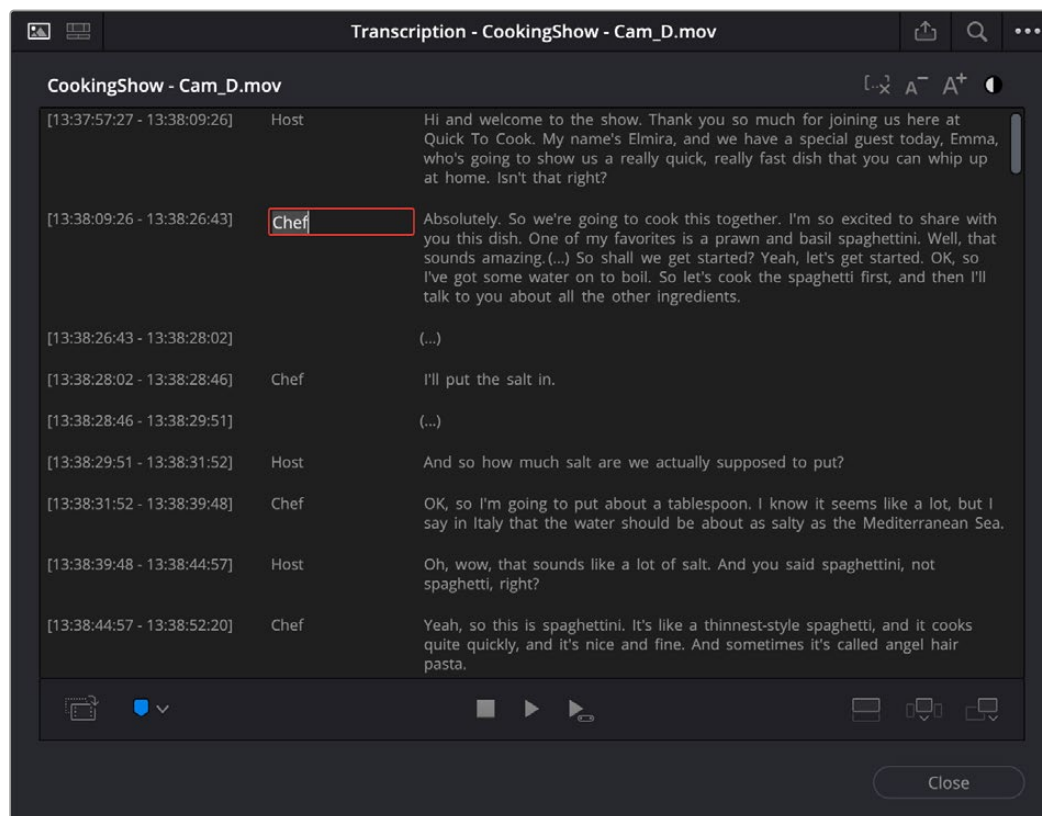
Turning on Speaker Detection in a clip's context menu.

To Enable Speaker Detection for Transcriptions:

- Right-click on a clip or timeline and select AI Tools > Audio Transcription > Speaker Detection.
- Or select Speaker Detection from the Transcription window's Option menu.
- Click on the Transcribe option to start the operation.

Transcription and Speaker Detection in Timelines

You can also choose to Transcribe and Detect Speakers for a timeline. This will analyze the timeline and transcribe the underlying media. It will then give you a timeline transcription based off the In and Out points of each clip in the timeline and assign speakers to each segment. Timeline Transcriptions are opened in Timeline Transcription mode, as described below.

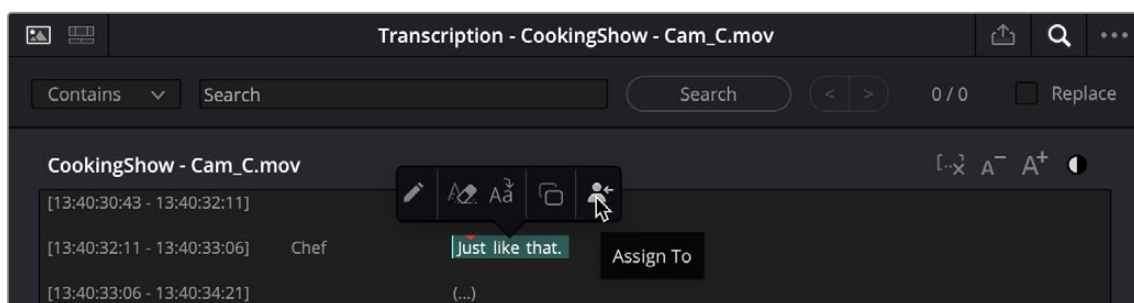


The Transcription window with Display Speakers active. You can change the name of a speaker by double-clicking the name and entering a new one in the text field.

With Speaker Detection enabled, any subsequent audio transcriptions you generate for clips and timelines will detect individual speakers. All the speaker's voice signatures are remembered at the project level across multiple clips. If you come back the next day to add more audio clips to your project, DaVinci Resolve will remember the voices you analyzed previously, and automatically assign them to the appropriate speaker. If you want to remove the saved speakers from your project, select Reset Speaker Data, from the Transcription window's Option menu.

In the Transcription window, speakers will initially be listed numerically (Speaker 1, 2, 3 etc.). You can change the name of the speaker by double-clicking on the speaker name and entering a new name in the text field. All other instances of that speaker will be changed to the new name.

If the Speaker Detection made an error about who is speaking, you can manually re-assign a text to another speaker by selecting the text, right-clicking on it and choosing the Assign To icon from the pop-up menu. Then another dialog box will appear letting you change the attribution to any of the other detected speakers.

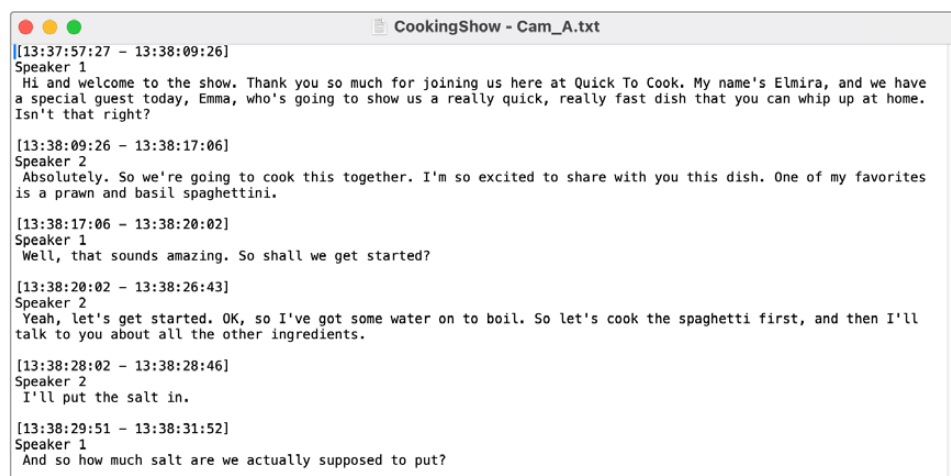


You can manually change the speaker attribution by right-clicking on the text and selecting the Assign To icon.

If you wish to hide the speakers again, in order to declutter the Transcription Window you can toggle their view on or off in the Transcription window's Option menu, under Display Speakers. Toggling this feature does not remove the speaker associations that have already been set.

Export Audio Transcriptions with Speaker and Timecode Information (Studio Version Only)

In DaVinci Resolve Studio when you export a transcription, the resulting text file will include speaker name and timecode information for each speech block. This useful information will make searching and navigating transcriptions much easier.



Transcripts include speaker and timecode information for dialog blocks.

Audio Transcription Supported Languages

You can choose which language is used to generate the transcription in the Project Settings > Subtitle and Transcription> Transcription Setup section. You can select any of the supported language models, or keep it on Auto for DaVinci Resolve to figure out who is speaking what.

As of this writing Audio Transcription and Text-Based Editing support the following languages:

- Danish, Dutch, English, French, German, Italian, Japanese, Korean, Mandarin (Simplified Traditional), Norwegian, Portuguese, Russian, Spanish and Swedish.
- Many other languages are also available by using the Extended AI Transcription Support Package in the Extras Downloads Manager. These include: Afrikaans, Arabic, Armenian, Azerbaijani, Belarusian, Bosnian, Bulgarian, Catalan, Croatian, Czech, Estonian, Finnish, Galician, Greek, Hebrew, Hindi, Hungarian, Icelandic, Indonesian, Kannada, Kazakh, Latvian, Lithuanian, Macedonian, Malay, Marathi, Maori, Nepali, Persian, Polish, Romanian, Serbian, Slovak, Slovenian, Swahili, Tagalog, Tamil, Thai, Turkish, Ukrainian, Urdu, Vietnamese, and Welsh.

IntelliScript for Creating Timelines from Scripts (Studio Version Only)

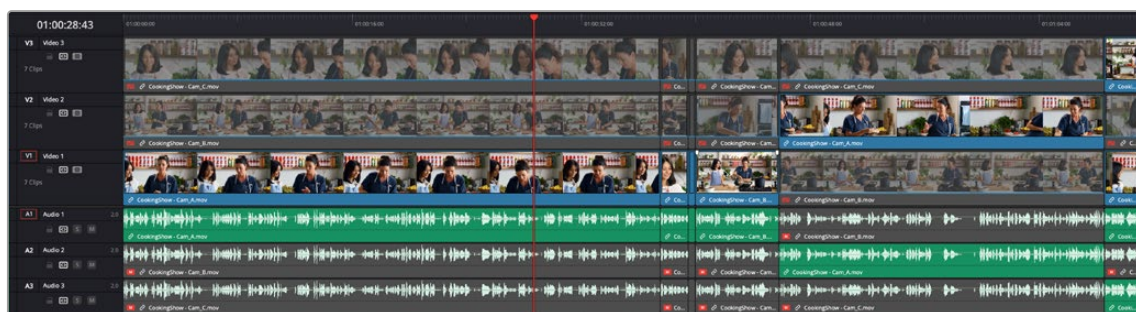
DaVinci Resolve 20 has a powerful AI-powered IntelliScript tool which uses the Transcription Engine to match and generate a timeline using an original script. This feature lets you quickly assemble rough cuts of scenes and even organizes multiple takes for you in the timeline. IntelliScript is smart enough to understand context, so if your actor ad-libs and uses a similar phrase rather than the exact phrase written in the script, it should still find it.

A simple text file script to feed into IntelliScript

To Use IntelliScript to generate a timeline based on a script:

- 1 Create a script file in plain text (.txt) format of just the spoken text. IntelliScript requires just the plain spoken text that has full stops (.) after sentences. Standard script formatting with scene descriptions, parentheticals, and character names will likely cause transcription issues.
- 2 Select all the clips you want to use for IntelliScript, right-click, and select AI Tools > Create New Timeline Using IntelliScript.
- 3 DaVinci Resolve will then prompt you to load the text script file. Press Open.

When the transcription process is done, IntelliScript will create a new timeline, and on track one, lay out the best clips in order based on the dialog in the script file. Any alternative takes are automatically placed on additionally created tracks that are disabled by default. Once the timeline is set up, you can review the best take on track one and preview alternative takes for specific regions, enabling and disabling them as needed.



The IntelliScript results in the timeline. It places the best takes on track one, alternative takes on separate tracks, and disables them by default. Here we overrode its choices at the end and manually enabled what take we thought was best.

NOTE: IntelliScript chooses the “best” take by matching the spoken dialog only. It does not use any good take metadata or make any judgments based on anything other than dialog matching.

Text-Based Video Editing (Studio Version Only)

After a Transcribe command has been applied, DaVinci Resolve now knows the exact timecode position of each word in the clip, and that information opens up several useful text-based editing tools using the Transcription window. Text-based video editing is available in the both the Cut and Edit pages.

The cursor in the Transcription window and the playhead in the Viewer are linked. As you move the cursor in the text, the playhead updates to that respective position in the clip in the Viewer. This allows you to precisely place the playhead exactly where you want it based on the text word, rather than having to scrub back and forth through the audio track.

To move the playhead to a specific word in the clip:

- Click on a word in the Transcript window, and it will highlight with a red arrow. The viewer playhead will then jump directly to the end of that word in the clip in the Viewer.

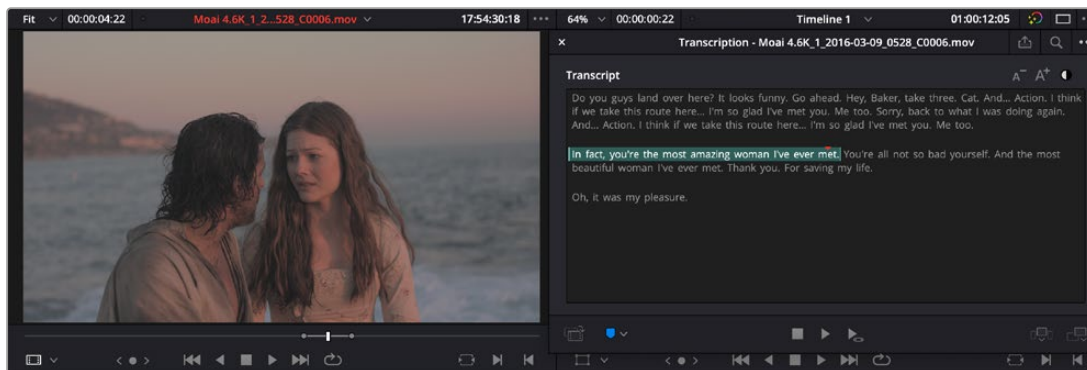
In addition to the cursor and playhead being linked, the In and Out points of a clip are also linked to the selected range of the text in the Transcription window.

To select an In and Out range in the clip:

- Click and drag over the text in the Transcription window to highlight it in teal. The In point of the clip is set just before the first word, and the Out point is set just after the last word in the highlighted text.

To select the In and Out range of a sentence:

- Click on any word in the sentence and press the X key to select the whole sentence and highlight it in teal.



Selecting a text range in the Transcription window (R) automatically sets In and Out points for that range on the clip in the Viewer (L).

With an In and Out range set, you can now use the editing tools in the Transcription window to perform some common editing and media management functions.



The Create Subclip and Add Marker tools in the Transcription window

To create a new subclip from a selected range:

- 1 Select a range of text in the Transcription window, by clicking and dragging to highlight the text in teal.
- 2 Press the Create Subclip icon in the lower left of the Transcription window.
- 3 In the Create Subclip window, enter a new name for the subclip and select the Create button.

To add a marker to a clip:

- 1 Either select a specific word in the Transcription window (red arrow) for a standard marker, or select a range of text by clicking and dragging to highlight the text in teal for a duration marker.
- 2 Click on the Marker icon in the lower left of the Transcription window.



The Place on Top, Insert, and Append Edit icons in the Transcription window

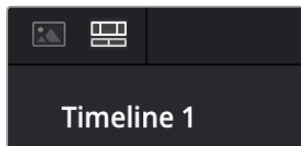
To edit an In and Out range into the current timeline:

- 1 Select a range of text in the Transcription window by clicking and dragging to highlight the text in teal.
- 2 Click one of the Edit icons in the lower right of the Transcription window.
 - **Place on Top:** Puts the selected edit range on a track above the one currently selected in the timeline.
 - **Insert Edit:** Inserts the selected edit range into the timeline at the timeline In point.
 - **Append:** Adds the selected edit range to the end of the timeline.

Edit Timelines Based on Source Clip Transcription

You can use the transcription you generated from the original source clip to subsequently modify that clip on the timeline. Entering this mode in the Transcription window will let you select, cut, and paste text, and then mirror those changes to the timeline.

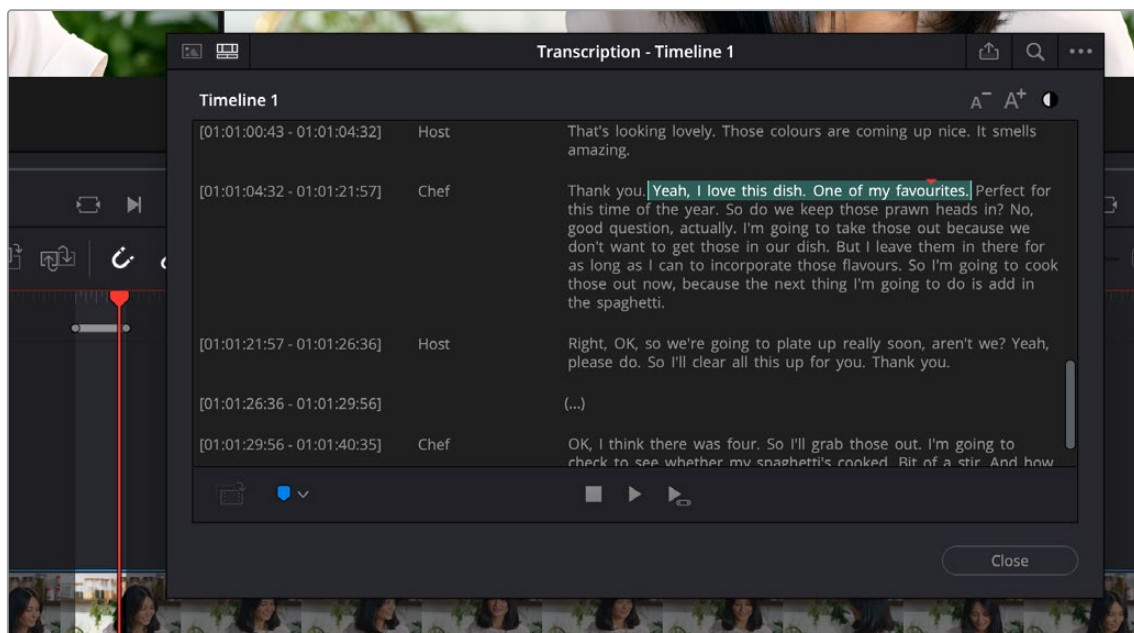
To use Timeline mode in the Transcription window, the clips on the timeline must be transcribed in the Media Pool. To enter Timeline Transcription mode, click on the Timeline icon in the upper left of the Transcription window.



The Timeline Transcription Mode icon in the upper right of the Transcription window

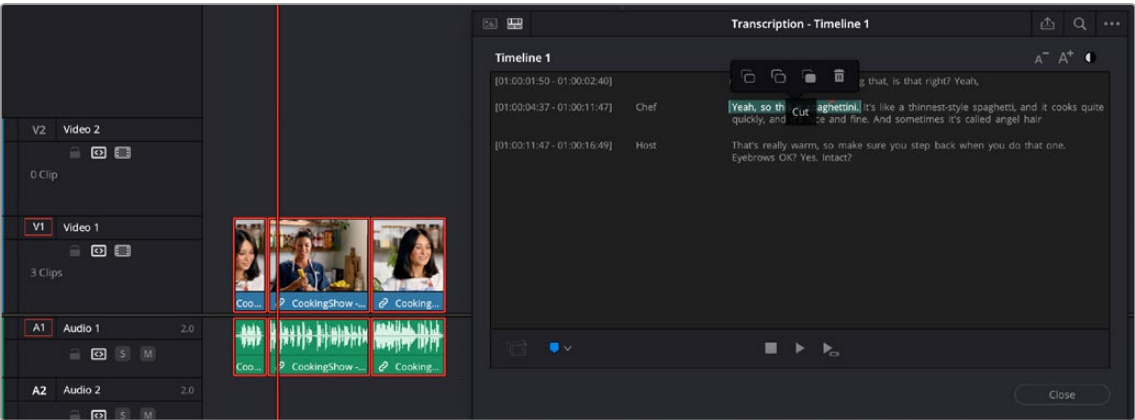
Once you've activated the Timeline Transcription mode, you can use the text editing tools in the window, and those changes will also appear on the timeline.

For example, in the clip below we selected a sentence in the Transcription window, and as you can see, the In and Out points of that sentence were automatically set in the Timeline behind it.

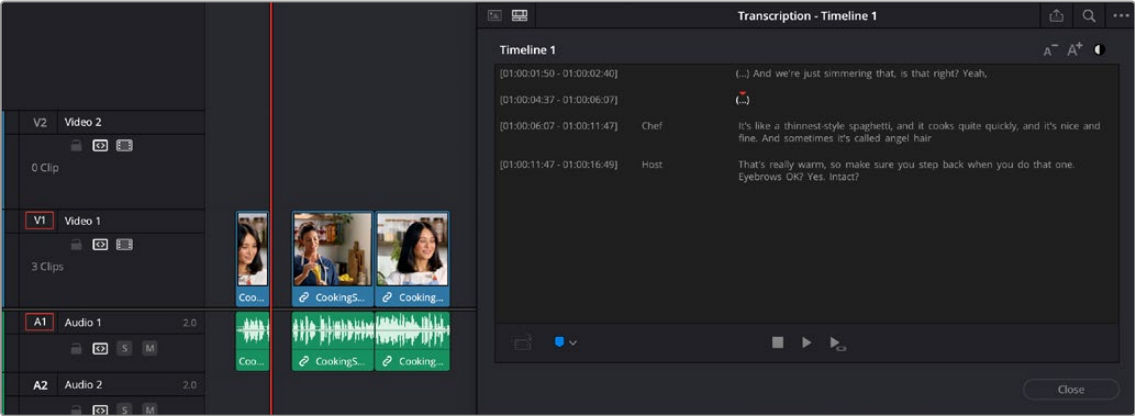


Selecting a sentence in the Transcription window sets the corresponding In and Out points in the timeline.

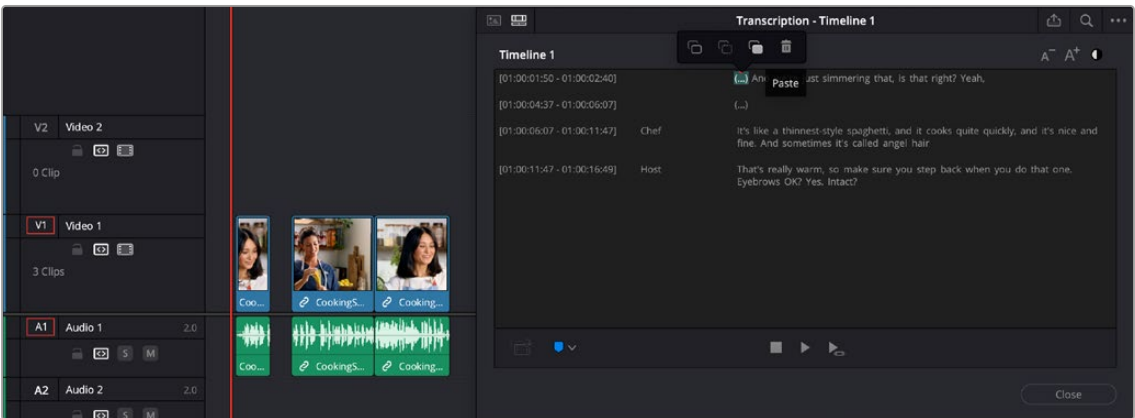
Once you have the text selected, you can manipulate that selection using the standard Cut, Copy, Paste, and Delete tools, and mirror those changes to the timeline. In the example below we will chop out one sentence of a clip, then move it to an earlier position in the timeline.



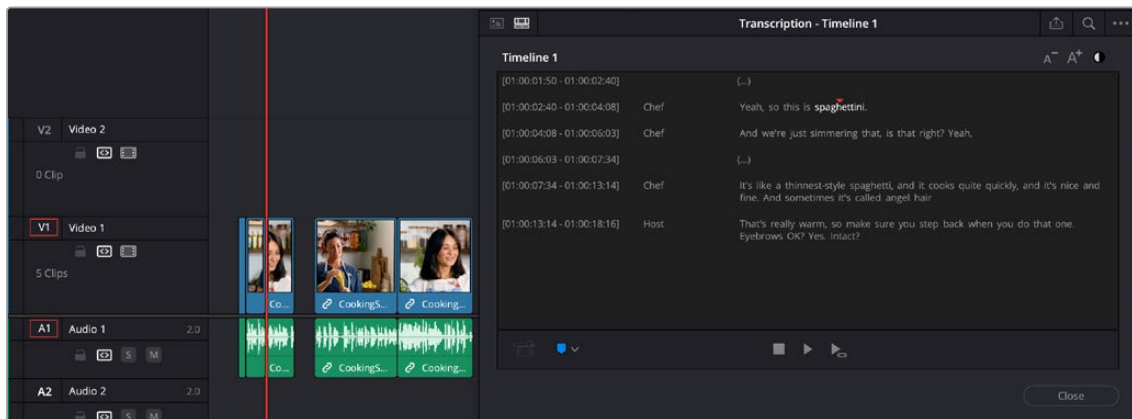
Selecting the first sentence of the second clip and then right-clicking and using the Cut tool



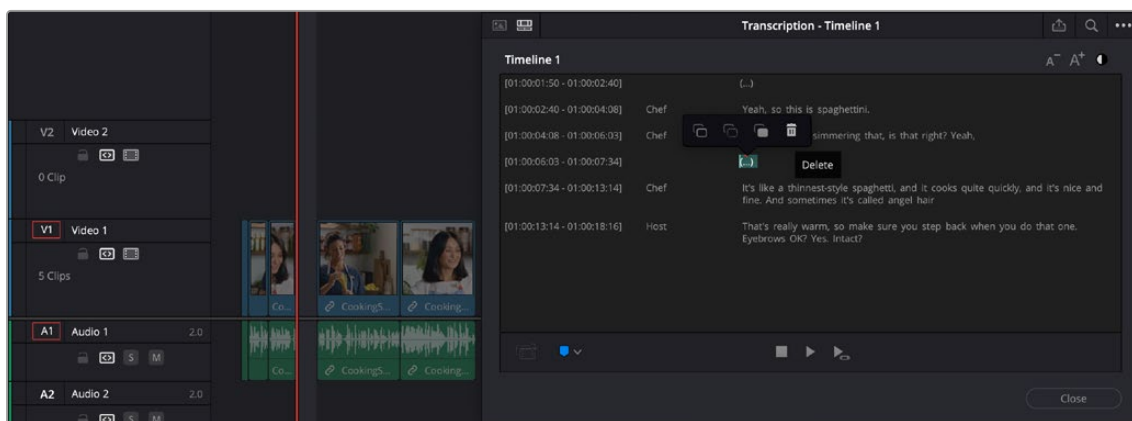
Here we see the sentence cut out of the timeline and replaced by silence (...) in the Transcription window.



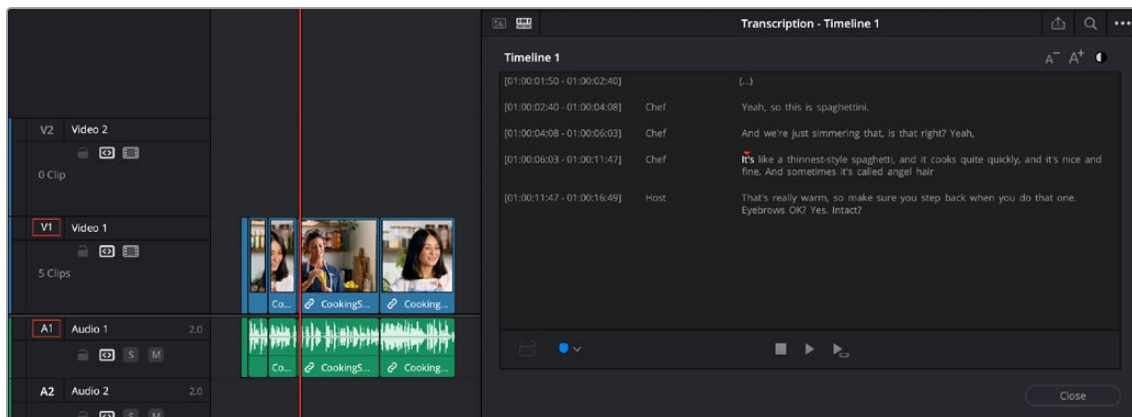
Now we select the new In point of in the timeline by selecting the first part of the sentence of the first clip in the Transcription window. Then we right-click and hit the Paste command.



The video of the sentence we cut out is pasted in that spot and now comes first in the Transcription window.



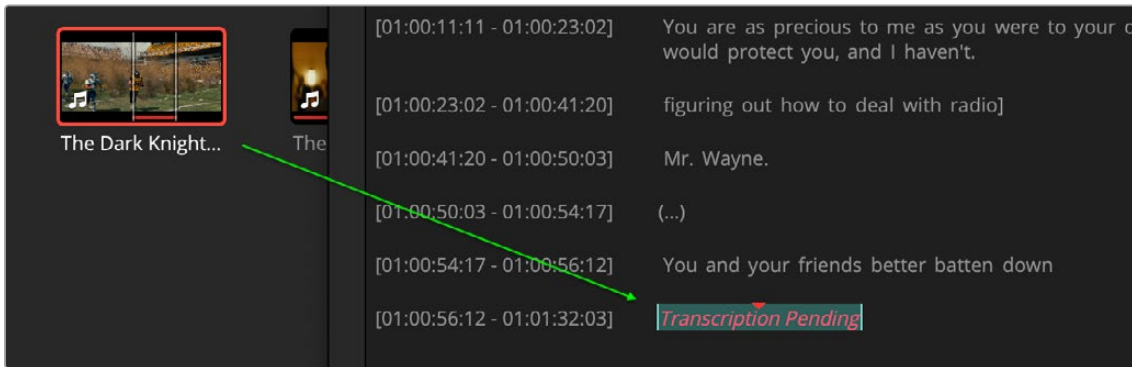
Now we select the hole in the timeline where we cut out the sentence originally by selecting the (...) silence in the Transcription window. Then right-click and choose Delete.



The final timeline with the sentence moved and the hole deleted. All done just by manipulating the text in the Transcription window.

Adding Clips to Timeline Transcriptions

While you are editing you can continue to add un-transcribed clips into the timeline. When an un-transcribed clip is detected, it is flagged in the Transcription window as “Transcription Pending” in this case, simply select “Transcribe Missing Clips in Timeline” from the Option menu to start the transcription of the all missing clips.



When an un-transcribed clip is added to the timeline, it's flagged Transcription Pending in the Transcription window

Import and Export Transcriptions using SRT Files

Transcription text can be imported and exported using standard .srt files.

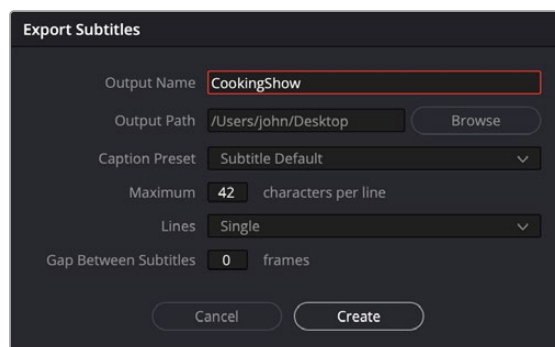
To Import a Transcription using an SRT File:

It's important that the timecode of the clip you're importing subtitles to, matches the timecode in the .srt file for accurate results.

- 1 Click on the Option menu in the Transcription window and select Import Subtitles from the dropdown menu.
- 2 Locate the .srt subtitle file in your file browser and click Open.
- 3 Click on the Transcribe icon in the Media Pool to open the Transcription window to view and edit the subtitles.

To Export a Transcription using an SRT File:

- 1 Click on the Option menu in the Transcription window and select Export Subtitles from the drop-down menu.
- 2 The Export Subtitles dialog box will appear.
- 3 Enter a file output name and path, to save your .srt file to.
- 4 Enter the subtitle information for Caption Preset, Maximum Characters per Line, Number of Lines, and Gap Between Subtitles, as needed.
- 5 Click on the Create button to save the .srt file.



The Export Subtitles dialog box from the Transcription window's Option menu.